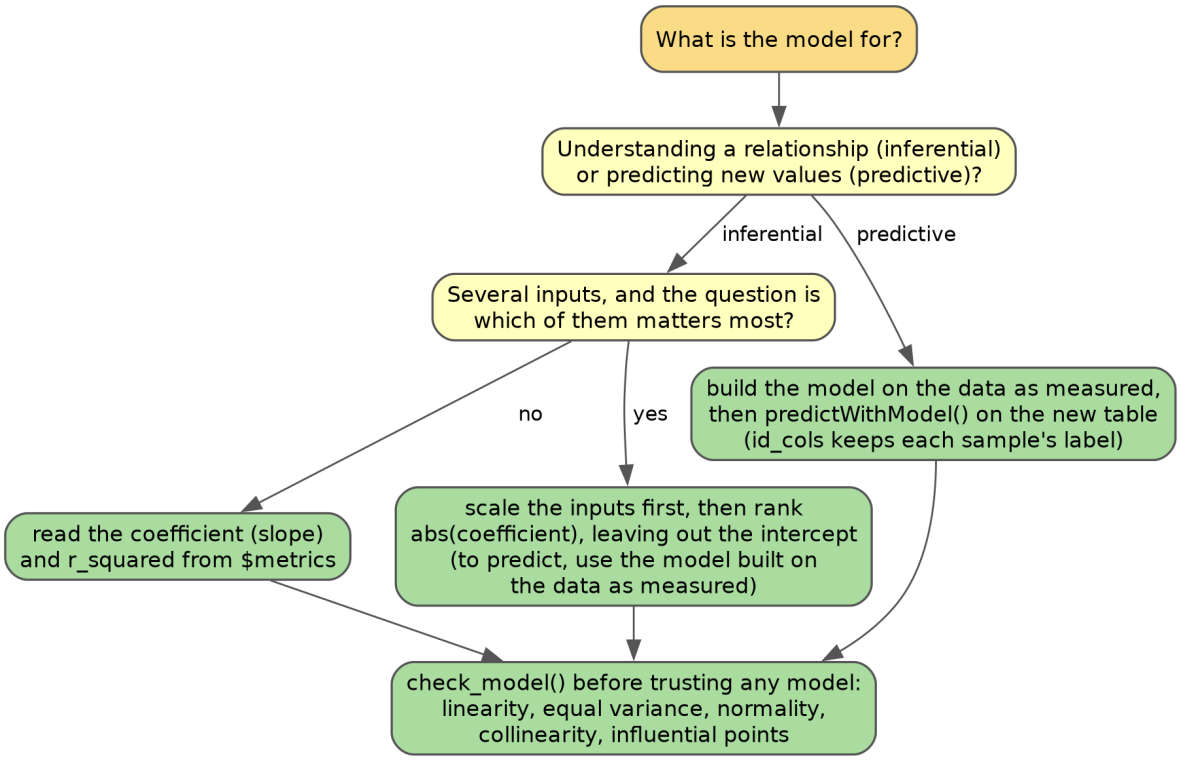
regression models: summary

Integrated Bioanalytical Chemistry · thebustalab.github.io/integrated_bioanalytics



Building a regression model starts with determining what the model will be used for. An **inferential** use reads the model's inner workings (the coefficients, their p-values, and r-squared) to describe how the output relates to the inputs. A **predictive** use applies the model to a table whose output was never measured. Three decisions follow. **How many inputs:** more inputs can explain more of the variation in the output, which r-squared shows. **Whether to scale:** scale the inputs before ranking them by the size of their coefficients, but predict with the model built on the data as measured. **Whether to trust it:** check the residuals, the inputs, and any influential points with `check_model()`.

Function or step	What it is for	What we set or vary
`Hmisc::impute()`	filling in missing values, because `buildModel()` stops on any	the fill value (here each column's median)
`buildModel()`	fitting a model	`data`, `model_type = "linear_regression"`, `input_variables` (one column or several), `output_variable`
`$model`, `$metrics`	the fitted model, and a table of its r-squared, coefficients and p-values	`filter(type == "coefficient")`, `filter(type == "statistic")`, `filter(variable == "r_squared")`
`predictWithModel()`	predictions for any table with the same input columns, returned as that table with a new `predicted_<output>` column	`data`, and `model` (the whole `buildModel()` output)
`mutate(across(c(...), scale))`	putting inputs on a mean of zero and a standard deviation of one before comparing coefficients	which inputs to scale
`arrange(desc(abs(value)))`	ranking coefficients by size	drop the intercept first with `variable != "(Intercept)"`
`check_model()`	diagnostic panels: posterior predictive check, linearity, homogeneity of variance, normality of residuals, collinearity, influential observations	given `model$model`